CABLE: a linear model that outperforms state-of-the-art EEG-to-fMRI models on new subjects

Summary. Simultaneous EEG-fMRI shows that scalp rhythms carry information about slow blood-oxygen-level-dependent (BOLD) activity throughout the brain, but BOLD-EEG interpolating models transfer poorly to new subjects. State-of-the-art models are deep networks: NeuroBOLT fine-tunes a pretrained EEG foundation model, and MEL feeds band-power tokens to neural-network heads; both predict only fMRI from EEG. We introduce CABLE (Causal Alignment of Band-Limited EEG), a linear model whose key step aligns each subject's EEG to itself: within each frequency band, EEG is whitened by that subject's running spatial covariance, computed only from past samples, so a new subject needs no fMRI, labels or calibration. An ensemble of ridge regressions reads out seven regional BOLD signals, including thalamus and putamen. In a prespecified evaluation on the NeuroBOLT dataset (eyes-closed rest; 22 subjects), with all three models run on identical subject-disjoint splits, CABLE reached r = 0.41, compared with 0.30 for NeuroBOLT (+37%) and 0.35 for MEL (+18%); leaving one subject out, it beat NeuroBOLT for 22/22 subjects and MEL for 18/22. On an untouched external dataset, OpenNeuro ds003768 (33 adults; different site, scanner and electrode cap), CABLE again led (0.37 vs 0.28 for NeuroBOLT, better for 31/33 participants, and 0.32 for MEL), and trained only on the NeuroBOLT dataset it reached 0.30 there, matching NeuroBOLT trained on ds003768 itself (0.28). The same projection also runs in reverse, predicting held-out subjects' slow EEG band-power fluctuations from regional fMRI (median r = 0.34, chance 0.00; 0.45 for alpha power, an index of vigilance). We also provide computational cost improvements: CABLE trains in CPU-minutes, while NeuroBOLT needs GPU-hours. Nonlinear heads and foundation-model features added at most 0.01, while per-subject alignment produced the largest single gain: thus, we prove EEG-BOLD mapping shared across subjects is almost linear, and the remaining error lies in subject-specific EEG-BOLD mappings.

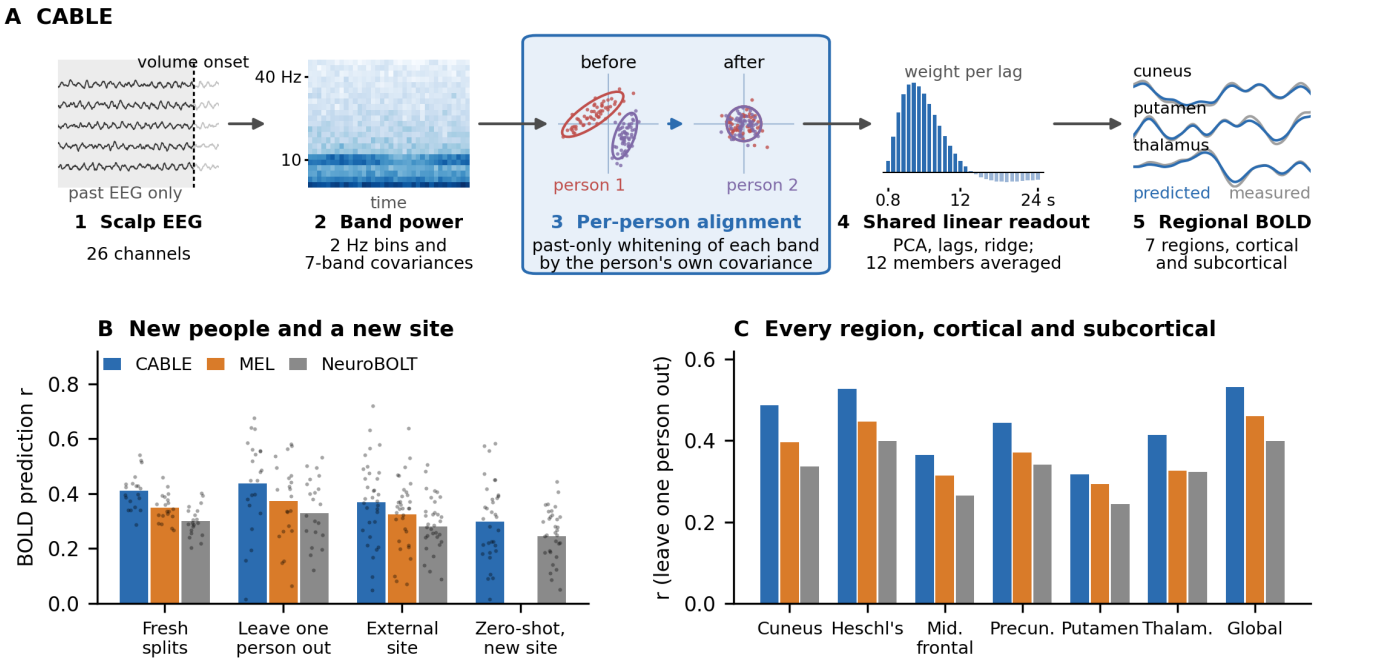


Figure 1. A CABLE (schematic; traces illustrative). Step 3 aligns each subject's EEG to itself using past samples only, so one linear projection encompasses all subjects. **B** Prespecified comparisons on new subjects and a new site. Bars represent mean r. Dots represent individual splits, such as held-out subjects (NeuroBOLT dataset) or external participants (ds003768, including zero-shot). CABLE beat NeuroBOLT for 22/22 held-out subjects (+0.110, 95% confidence interval [0.087, 0.133]) and 31/33 external participants; every comparison shown has $p < 0.001$ (two-sided paired sign-flip tests over subjects, also for the fresh splits, which reuse the 22 subjects). Zero-shot: trained on the NeuroBOLT dataset only, tested on ds003768; MEL was not evaluated zero-shot. **C** Accuracy per ROI, leaving one subject out (22 subjects, mean over 29 scans).

CABLE. From 26 scalp channels, CABLE computes log power in 2 Hz bins (1 to 45 Hz) and spatial covariances in seven frequency bands on a 0.8 s grid, using only EEG recorded before each time point. Its central step aligns each subject's EEG to itself: within each frequency band, the signal is whitened by that subject's running spatial covariance [1], removing subject-specific channel scaling and mixing, and band covariances are re-centered on

their running mean [2]. These statistics accumulate from the start of the recording, so alignment needs no fMRI or labels from a new subject. Aligned features are reduced by principal components, expanded with hemodynamic lags spanning the preceding 0.8 to 24 s, and mapped to seven regional BOLD signals by ridge regression. Twelve members with different feature sets (eight aligned, four unaligned) are averaged (Fig. 1A). Hyperparameters are set by nested cross-validation within the training set; a perturbation test confirms that no prediction uses later EEG.

Evaluation. The main dataset is the public simultaneous EEG-fMRI data released with NeuroBOLT [3] (Vanderbilt; eyes-closed rest; 29 scans, 22 subjects), with its seven regional targets (Fig. 1C). The protocol was prespecified and dated before scoring: 20 newly drawn subject-disjoint train/test splits, leave-one-subject-out cross-validation, and an untouched external dataset, OpenNeuro ds003768 [4] (simultaneous EEG-fMRI from another site, scanner and electrode cap; 33 adults; preprocessed with open-source tools). NeuroBOLT was run from its released code without modification and MEL [5] was reimplemented from its paper, as it has no public code. CABLE was designed on 17 of the 22 subjects; on the 5 never used in design its lead was unchanged (+0.11 over NeuroBOLT, 5 of 5; +0.06 over MEL, 4 of 5), and ds003768 was untouched until the method was frozen.

Results. Leaving one subject out on the NeuroBOLT dataset, CABLE reached 0.438, compared with 0.328 for NeuroBOLT and 0.372 for MEL. Trained on the NeuroBOLT dataset and tested on ds003768, CABLE reached 0.298, compared with 0.245 for NeuroBOLT trained the same way (better for 27 of 33 participants), and matched NeuroBOLT trained on ds003768 itself (0.280; difference +0.021, 95% CI [−0.001, 0.042]; Fig. 1B).

And in reverse. NeuroBOLT and MEL predict only fMRI from EEG. The same lagged ridge projection, without the alignment step, also runs backwards: it maps 64 regional fMRI signals plus global signals (20 components, lags 0 to 16 s) to the slow (0.01 to 0.15 Hz) power envelopes of a held-out subject's EEG rhythms (26 channels, 5 bands), the timescale that BOLD sampled every 2.1 s can resolve. Leaving one subject out, median r was 0.34 (95% CI [0.29, 0.39]; chance from circularly shifted predictions, 0.00), and 0.45 for scalp-mean alpha power, an index of vigilance. The information is regional: removing the global signals left both unchanged, and the global signals alone reached only 0.29 and 0.39.

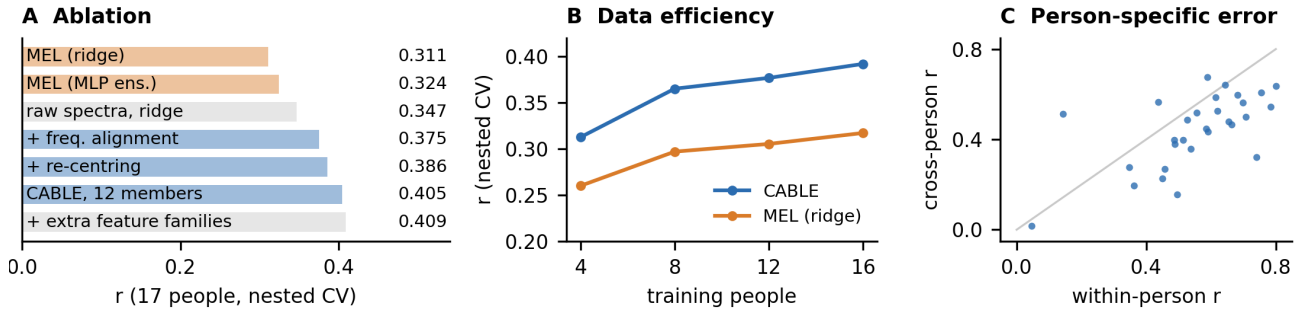


Figure 2. A Development ablation (17 subjects, nested cross-validation; MEL in the same folds). **B** Data efficiency in development (17 subjects; MEL-Ridge only): with 4 training subjects CABLE matches MEL-Ridge trained on 16. **C** Each dot is one scan. Horizontal: r with earlier data from the same scan in training (NeuroBOLT's intra-scan protocol; 48 scored volumes starting about 2 min after the training data, which limits autocorrelation; noisy); vertical: r over the whole scan when the subject is never seen. Points well below the diagonal: BOLD predictable from the subject's own EEG but not through the mapping learned from others.

Why CABLE works

Alignment is the largest single gain. In development, ridge on raw spectra already reached r = 0.347, above both MEL variants (0.311 and 0.324); aligning each subject's EEG raised it to 0.375 (frequency-specific alignment) and 0.386 (with re-centring), and the full ensemble reached 0.405 (Fig. 2A). Nonlinear capacity added little: multilayer perceptrons, gradient-boosted trees and frozen embeddings from LaBraM (the foundation model that NeuroBOLT fine-tunes [6]) each improved on ridge by at most 0.01. At this data scale, the EEG-BOLD mapping shared across subjects is almost linear; the dominant difference between subjects lies in EEG statistics, which alignment removes.

The remaining error is subject-specific. Accuracy per subject ranges from r = 0.02 to 0.68 on leave-one-subject-out tests. Several subjects whose EEG predicts their own BOLD well (r=0.45-0.50 within-scan) are predicted poorly by the shared mapping (r=0.16-0.27; Fig. 2C). A per-scan delay does not explain this: in development, an oracle per-scan hemodynamic delay recovers only 0.02.

[1] He & Wu, IEEE TBME 2019. [2] Zanini et al., IEEE TBME 2018. [3] Li et al., NeuroBOLT, NeurIPS 2024. [4] OpenNeuro ds003768. [5] Liu et al., MEL, arXiv:2608.29304. [6] Jiang et al., LaBraM, ICLR 2024.